

TORONTO PEARSON, ON, Canada

WMO: 716240

Lat: 43.6658N

Lon: 79.6062W

Elev: 173

StdP: 99.26

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.5	-15.7	-24.2	0.4	-17.9	-21.0	0.6	-14.9	14.2	-5.5	12.7	-4.7	5.3	0	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	31.5	22.4	29.7	21.5	28.1	20.8	23.8	29.2	22.8	28.0	21.9	26.5	6.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.1	17.2	26.8	21.1	16.1	25.7	20.2	15.2	24.8	72.2	29.2	68.3	28.2	64.9	26.6	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.7	9.4	-21.7	34.1	3.5	1.9	-24.3	35.4	-26.3	36.6	-28.3	37.7	-30.9	39.1		
(5)			-22.2	25.7	3.4	1.2	-24.7	26.5	-26.7	27.2	-28.6	27.9	-31.0	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-5.0	-4.1	0.5	7.0	13.8	19.4	22.1	21.3	17.4	10.5	4.2	-1.2		
	DBStd	10.63	6.33	5.62	5.67	4.64	4.55	3.79	3.14	3.00	4.03	4.40	4.58	5.02		
	HDD10.0	1859	464	395	298	112	13	0	0	0	1	47	182	347		
	HDD18.3	3779	722	628	552	339	154	32	4	8	64	245	425	606		
	CDD10.0	1460	0	0	4	22	132	281	376	351	223	63	6	1		
	CDD18.3	339	0	0	0	1	15	63	122	100	36	3	0	0		
	CDH23.3	2501	0	0	1	8	132	477	959	667	242	15	0	0		
	CDH26.7	702	0	0	0	1	28	143	299	168	61	2	0	0		
	WSAvg	4.4	5.0	5.0	4.9	4.9	4.2	3.9	3.9	3.7	3.8	4.2	4.5	4.8		
	PrecAvg	804	58	54	54	74	71	79	79	74	65	69	66	62		
(15) Precipitation	PrecMax	1025	129	109	110	130	130	170	177	139	164	133	127	114		
	PrecMin	597	24	28	21	27	10	30	20	17	18	19	21	25		
	PrecStd	117	26	23	26	28	28	40	39	31	31	32	28	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	12.0	19.8	24.3	29.5	32.7	34.1	33.2	31.6	25.4	17.6	13.0		
		MCWB	10.7	8.3	13.6	15.5	20.7	22.9	23.9	23.1	22.4	18.5	13.4	10.5		
	2%	DB	7.9	7.7	14.9	20.2	27.0	30.4	31.7	30.2	28.3	22.1	15.1	9.5		
		MCWB	6.2	5.1	10.3	13.2	18.8	21.6	22.7	22.2	21.0	17.0	12.1	7.6		
	5%	DB	5.2	5.0	11.2	17.2	24.3	28.1	29.9	28.4	26.1	19.6	13.1	7.1		
		MCWB	3.9	3.0	7.6	11.0	17.4	20.7	21.6	21.0	19.9	15.5	10.4	5.4		
	10%	DB	3.2	3.1	8.1	14.5	21.6	26.0	28.1	26.9	24.0	17.3	10.9	5.1		
		MCWB	1.9	1.4	5.0	9.2	15.6	19.2	20.8	20.3	18.8	13.9	8.5	3.5		
	0.4%	WB	10.8	8.5	14.9	17.1	22.1	24.1	25.7	24.7	23.6	20.5	14.6	11.1		
		MCDB	11.7	10.8	18.8	21.8	28.2	30.7	31.7	29.3	29.1	23.5	16.3	12.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	6.5	5.5	11.2	14.6	20.1	22.8	24.1	23.3	22.2	17.6	12.6	8.0		
		MCDB	7.5	7.4	14.0	18.9	25.6	28.9	29.3	28.3	26.8	21.0	14.9	9.3		
	5%	WB	4.0	3.2	8.0	12.0	18.2	21.5	22.9	22.4	20.9	16.0	10.6	5.7		
		MCDB	5.1	4.7	10.7	16.1	23.1	26.6	28.0	26.9	24.5	18.8	12.7	7.0		
	10%	WB	1.9	1.5	5.3	9.7	16.3	20.3	21.8	21.4	19.5	14.5	8.8	3.6		
		MCDB	3.0	3.0	7.8	13.5	20.7	24.7	26.4	25.4	23.0	17.0	10.9	5.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.3	8.0	9.2	9.7	9.4	9.5	9.2	9.3	8.0	6.8	6.0		
		MCDBR	8.0	8.9	12.1	13.8	12.4	11.8	11.2	10.8	11.5	10.9	9.8	7.8		
		MCWBR	7.5	7.6	8.9	9.3	7.3	6.3	5.4	5.2	6.0	7.0	7.8	6.9		
		MCDBR	7.8	8.5	11.7	12.2	11.7	10.6	9.7	9.4	9.8	10.0	9.3	7.6		
	5% WB	MCWBR	7.7	7.8	9.3	9.4	7.4	6.5	5.4	5.2	5.9	7.1	8.0	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.294	0.320	0.331	0.376	0.404	0.434	0.426	0.413	0.381	0.360	0.316	0.298		
	taud		2.395	2.310	2.398	2.311	2.283	2.225	2.270	2.296	2.366	2.407	2.490	2.434		
	Ebn at Noon		835	868	909	888	869	840	843	842	843	813	807	796		
	Edh at Noon		76	99	104	123	130	138	131	123	106	90	70	68		
(44) All-Sky Solar Radiation	RadAvg		1.42	2.17	3.45	4.43	5.37	5.85	5.86	5.14	4.12	2.47	1.54	1.12		
	RadStd		0.14	0.21	0.35	0.47	0.56	0.42	0.42	0.26	0.39	0.23	0.20	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+88	N/A
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page